ANDREA PASOTTI

Biomedical Engineer · Algorithms for Biosignals, Wearable Sensors and Medical Images

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PROFILE

Biomedical engineer with over 4 years of experience developing and validating algorithms for biomedical signals and images, mainly from wearable inertial sensors (IMUs), across clinical research (IRCCS Institute of Neurological Sciences of Bologna, University of Bologna) and medical device R&D under ISO 13485, ISO 14971 and the EU MDR. I build pipelines in MATLAB and Python, from preprocessing to quantitative analysis, and validate them against clinical references on real patient data.

EXPERIENCE

Research Fellow – Biomedical Signal Processing Algorithms Apr 2025 – present

University of Bologna, Department of Electrical, Electronic and Information Engineering (DEI) · Cesena, Italy

- Develop algorithms for quantitative monitoring of motor development in preterm infants from IMU signals: preprocessing and estimation of U-turns, push-off and step width with time, frequency and wavelet methods.
- Automated the segmentation of gait and U-turns, robust even in toddlers taking their first steps, where manual segmentation is slow and difficult; it now runs on the group's entire database of thousands of subjects.
- Validate results against clinical references and integrate the code into a digital twin framework; 2 contributions to SIAMOC 2026 and 2 articles in preparation for IEEE Transactions on Biomedical Engineering.

R&D Engineer, Medical Devices, and Quality Specialist Jun 2024 – Mar 2025

DEAS S.r.l. (respiratory medical devices) · Castel Bolognese, Italy

- End-to-end design of a vibrating-mesh nebulizer, from selection and characterisation of the piezoelectric actuator to verification; contributed to the design of a high-flow therapy device.
- Risk analysis (ISO 14971), functional, safety and flow verification protocols and reports for single-use devices and PEEP valves, as part of the technical documentation (EU MDR 2017/745, ISO 13485).
- Coordinated aerosol particle-size measurements with an external laboratory and analysed the results.

Biomedical Engineer – Clinical Movement Analysis Jul 2022 – Feb 2025

IRCCS Institute of Neurological Sciences of Bologna, Neurosurgery Unit · Bologna, Italy · *remote; from June 2024 in parallel with DEAS*

- Designed and built on my own the analysis pipeline of the ARCHIMEDE study: 84 patients assessed with 3 inertial sensors before spinal decompression surgery and at 30 and 90 days after it.
- Posture, range-of-motion and gait parameters, including variability, multiscale entropy and recurrence quantification analysis; statistical comparison between two surgical techniques and with healthy subjects.
- In the 40 patients with 90-day follow-up, stride variability decreases with both techniques but gait remains slower than in healthy subjects; 3 conference contributions as second author (ESMAC 2025, SIAMOC 2024 and 2025) and 2 articles in preparation for Clinical Biomechanics.

EDUCATION

MSc in Biomedical Engineering (eHealth) · 100/110 2019 – 2022

Politecnico di Torino, Italy

- Thesis: automatic segmentation of sagittal T1 brain MRI in 98 paediatric patients with Chiari malformation. I developed the algorithms and analyses: wavelet and bilateral-filter preprocessing, Demons atlas registration, GVF active contours, with Dice 0.83 (brain), 0.73 (cerebellum) and 0.75 (posterior fossa) against manual segmentations; SVM classification of surgical indication with 93.7% accuracy on a held-out test set.

BSc in Biomedical Engineering · 97/110 2016 – 2019

University of Bologna, Italy

ACADEMIC IMAGING PROJECTS

- **Optic nerve ultrasound:** MATLAB pipeline on 75 real transorbital ultrasound images (equalisation, filtering, Otsu thresholding, distance transform, active contours) for calibrated measurements of the optic nerve and sheath diameters.
- **Rectal cancer CAD:** segmentation and characterisation on T2 MRI of 39 patients, with 22 imaging features, clustering, class balancing, feature selection and classification.

TECHNICAL SKILLS

Languages and tools: MATLAB (expert), Python (NumPy, SciPy, pandas, scikit-learn, statsmodels, PyWavelets, pydicom, SimpleITK), Git

Signal and image processing: filtering and convolution, Fourier and wavelet analysis, IMU-based gait analysis, time-series analysis, segmentation (active contours, Demons registration), feature extraction, validation metrics against expert references

Data analysis: inferential statistics, supervised and unsupervised learning, feature selection, model validation

Medical devices: design, verification and validation, risk management, technical documentation; ISO 13485, ISO 14971, EU MDR 2017/745

Languages: Italian (native), English (C1)

CONFERENCE CONTRIBUTIONS

- **2026** · Quantification of push-off power during gait: an IMU-based energetic approach. Stagni R., **Pasotti A.**, Biserna D., Bisi M.C. SIAMOC 2026.
- **2026** · Can feet distance during gait be quantified using IMUs? An energy-based estimation approach. Stagni R., Biserna D., **Pasotti A.**, Bisi M.C. SIAMOC 2026.
- **2025** · An IMU-based ambulatory assessment for spinal decompressive surgery. Stagni R., **Pasotti A.** et al. Gait & Posture 121, 234–235 (ESMAC 2025). DOI 10.1016/j.gaitpost.2025.07.251
- **2025** · Lumbar spinal laminectomy with and without fixation: an IMU-based ambulatory follow-up assessment. Stagni R., **Pasotti A.** et al. SIAMOC 2025.
- **2025** · Early longitudinal monitoring of motor development in preterm infants: is gait initiation enough? Stagni R. et al. (**Pasotti A.**, 7th of 9 authors). Gait & Posture 122. DOI 10.1016/j.gaitpost.2025.08.062
- **2025** · Sensor-based gait analysis for clinical monitoring and assessing treatment response in Juvenile Myasthenia Gravis. Bisi M.C. et al. (**Pasotti A.**, 6th of 8 authors). Gait & Posture 122. DOI 10.1016/j.gaitpost.2025.08.015
- **2024** · IMU-based ambulatory pre-surgical assessment of spinal decompressive surgery. Stagni R., **Pasotti A.** et al. SIAMOC 2024.
- **In preparation:** 7 journal articles, 3 as second author (IEEE Transactions on Biomedical Engineering, Clinical Biomechanics).

OTHER ACTIVITIES

- Teaching assistant, Fundamentals of Biomechanics, University of Bologna (2022–2026, about 50 hours per year).
- Testing and release support for an iOS app promoting physical activity in cancer prevention, Erasmus+ project uCanACT, University of Bologna (2024–2025).